RECRUITER SNAPSHOT | LIS / CLINICAL APPLICATIONS PORTFOLIO

# 71% TAT spike. Chemistry was not the bottleneck.

## I validated the clinical transaction, localized the delay to transport, separated a second send-out workflow, and designed a controlled fix with production-safe validation.

| ALLEN STALCUP | ASCP MLT | TARGET ROLES | PORTFOLIO |
|---|---|---|---|
| **Portfolio case study** | **Laboratory domain** | **LIS Analyst \| Clinical Applications \| Lab Systems** | **github.com/stalcup-dev** |

### SIMULATED CASE OUTCOMES

| 49 -> 84 min | 18 -> 54 min | 72% -> 96% | 37 -> 16 min |
|---|---|---|---|
| Headline TAT symptom | Collection -> receipt | Send-out SLA after fix | Staging delay |

All metrics, messages, defects, interventions, and outcomes are synthetic and public-safe; no PHI is used.

### WHAT I DID

- Traced an HL7 v2 ORU transaction with message-control ID and ACK evidence.
- Validated patient, order, specimen, result, interpretation, status, and timestamp integrity.
- Decomposed TAT until the delay localized to a missed transport window.
- Separated microbiology send-outs from routine chemistry and isolated batch waiting.
- Defined negative tests, exception handling, canary deployment, monitoring, and rollback.

### WHAT THIS PROVES

- **LIS / EHR interface troubleshooting**
- **HL7 v2 message and ACK literacy**
- **Patient-safety and data-integrity thinking**
- **Master-data / test-code mapping controls**
- **Laboratory TAT root-cause analysis**
- **Production change-control discipline**

### RECRUITER TAKEAWAY

**I can bridge bench-lab domain knowledge and healthcare IT: trace clinical data end to end, find the first failed handoff, and recommend the smallest safe change.**

**DEEP-DIVE EVIDENCE INSIDE: HL7/ACK trace | silent CBCDIFF -> CBC mapping failure | TAT drill-down | critical-result integrity | change control + rollback**

Case disclosure: simulated, public-safe portfolio scenario created to demonstrate LIS/clinical-systems reasoning. It does not represent production work performed at a specific hospital.

TECHNICAL EVIDENCE | PROVE THE TRANSACTION BEFORE USING IT AS EVIDENCE

# A technically successful message can still be clinically wrong

A dashboard can be numerically clean and clinically false if the interface changes identity, order linkage, interpretation, status, or timing. So the investigation starts with transaction evidence, not with the chart.

```
MSH|^~\&|LAB_LIS|LAB|EHR|HOSPITAL|202608181451||ORU^R01|MSG12345|P|2.5
PID|1||12345^^^HOSPITAL_A^MR||SMITH^JOHN||19800102
OBR|1|ORD200|ACC200|K^Potassium|||202608181400||||||202608181423|||||||202608181451
OBX|1|NM|K^Potassium||6.8|mmol/L|3.5-5.1|HH|||F

ACK from receiver:
MSA|AA|MSG12345
```

| Evidence | Meaning | Analyst question |
|---|---|---|
| **MSH-10 = MSG12345** | Message control ID | Can I trace the same transaction through every interface hop? |
| **PID-3** | Patient identifier + authority/type | Did the namespace remain intact? |
| **OBR-2 / OBR-3** | Placer order / filler order | Did the correct order stay paired with the lab-side identifier? |
| **OBR-7 / 14 / 22** | Collection / receipt / final-report event time | Are the timestamp semantics and chronology valid? |
| **OBX-5 / 6 / 8 / 11** | Value / units / interpretation / status | Did clinical meaning survive? |
| **MSA\|AA\|MSG12345** | Receiver accepted the message | Was it accepted - and did it later file/display correctly? |

**IMPORTANT DISTINCTION**

An AA acknowledgment proves acceptance at the application/message layer; it does not, by itself, prove correct patient filing, downstream workflow, or clinician display. The investigation still follows the result to its final destination.

## Critical-result integrity is not the same as critical-result notification

If the LIS classifies potassium 6.8 as critical high and the outbound result carries HH, preserving HH is a data-integrity control. The organization still needs a separate, policy-defined critical-result notification workflow - such as escalation, documentation, read-back, or acknowledgment where applicable. A perfect HL7 flag does not prove that operational notification occurred.

## Silent failure example: the interface works, the clinical intent does not

| Stage | Expected | Silent failure |
|---|---|---|
| **EHR order code** | CBCDIFF | CBCDIFF |
| **Interface crosswalk** | CBCDIFF -> CBCD | CBCDIFF -> CBC |
| **LIS receives** | CBC with differential | CBC only |
| **Technical status** | Message accepted | Message accepted |
| **Clinical outcome** | CBC + differential | Differential omitted |

**MASTER-DATA CONTROL**

Code crosswalks need an owner, source code, target code, display meaning, expected components, version/effective date, and test evidence. A mapping can be syntactically valid and clinically unsafe.

INVESTIGATION | FOLLOW THE 35 MINUTES UNTIL THE DELAY HAS AN OWNER

# The first cut breaks the original assumption

Once patient identity, order relationships, result meaning, and event timestamps are believable, I decompose the headline TAT. The chemistry interval is essentially unchanged. The deterioration is upstream.

| Interval | Last month | This month | Interpretation |
|---|---|---|---|
| **Collection -> received** | 18 min | 54 min | +36 min: deterioration |
| **Received -> final** | 31 min | 30 min | Stable analytical interval |
| **Collection -> final** | 49 min | 84 min | Headline symptom |

**DECISION**
Do not redesign chemistry. The evidence moves the investigation into collection, transport, and receipt.

## The second cut localizes the problem to one site

| Location | Last month | This month | Read |
|---|---|---|---|
| **Emergency Department** | 12 min | 13 min | Stable |
| **Inpatient floors** | 16 min | 17 min | Stable |
| **Main outpatient clinic** | 21 min | 22 min | Stable |
| **Satellite Clinic** | 24 min | 96 min | Localized deterioration |

## The third cut finds the failure boundary

| Drill-down | Last month | This month | What it tells me |
|---|---|---|---|
| **Collection -> pickup** | 10 min | 11 min | Clinic handoff is essentially stable |
| **Pickup -> lab receipt** | 14 min | 85 min | Transport becomes the failure boundary |

The average still does not tell the whole story. Most transports remain 15-22 minutes; a smaller group takes 150-210 minutes. The severe cases cluster around the Tuesday and Thursday 13:30 pickup window. On those days, the scheduled run is missing and specimens wait for the next available transport.

**EVIDENCE BEFORE OWNERSHIP**
I would verify schedule, pickup scan, route/driver, and arrival evidence before assigning root cause. The pattern creates the hypothesis; the operational record proves or disproves it.

## Contain the risk first

Cover the Tuesday/Thursday 13:30 pickup window immediately and define an escalation if the scheduled pickup does not occur. Then determine why the run is being missed - schedule design, staffing, route conflict, handoff, or documentation - and correct that mechanism rather than permanently adding resources without evidence.

SECOND WORKFLOW | THE FIRST FIX WORKS - THEN ANOTHER PROCESS REVEALS ITSELF

# Separate the population before judging the process

After the missed courier runs are contained, most routine chemistry specimens normalize, but several >60-minute delays remain. They all belong to microbiology send-outs. They are not random chemistry outliers; they follow a different workflow and need a different SLA and process model.

| Micro send-out step | Minutes | Interpretation |
|---|---|---|
| **Order verification** | 4 | Required work |
| **Required processing** | 6 | Required work |
| **Packaging** | 3 | Required work |
| **Staff availability** | 2 | Minor wait |
| **Batch waiting** | 22 | Primary avoidable delay |
| **Total main lab -> staging** | 37 | 22 min is waiting, not processing |

## The operational decision

The 30-minute batching rule reduces staff trips but makes the service level unreliable. Moving every specimen immediately would maximize speed at unnecessary labor cost. I would reduce batching to 10 minutes, keep an exception pathway for urgent specimens, and then verify both SLA performance and labor burden before making the rule permanent.

| Metric | Before | After | Decision signal |
|---|---|---|---|
| **Send-out SLA compliance** | 72% | 96% | Major outcome improvement |
| **Main lab -> staging** | 37 min | 16 min | Avoidable waiting reduced |
| **Staging trips / day** | 12 | 30 | Higher movement cost |
| **Overtime** | Baseline | No meaningful change | No demonstrated labor penalty |

**WHY KEEP THE 10-MINUTE RULE?**
Because the controlled objective improved materially and the added movement has not produced a demonstrated labor penalty. Do not optimize trip count merely because 30 sounds high. Change again only when workload evidence shows a real burden.

## What the story actually demonstrates

The value is not the synthetic percentages. It is the method: verify the transaction, decompose the workflow, localize the first changed stage, separate process populations, contain patient/operational risk, make the smallest defensible change, and prove that the change holds.

| Stage | Question | Decision |
|---|---|---|
| **Detect** | What changed? | Overall TAT increased |
| **Validate** | Can I trust the transaction? | Identity/meaning/time must pass |
| **Localize** | Where did time enter? | Transport, not chemistry |
| **Root cause** | Which event explains the tail? | Missed courier run |
| **Segment** | Are remaining cases the same process? | No - micro send-outs |
| **Correct** | What is the smallest safe change? | 10-minute batching |
| **Verify** | Did outcome improve without new harm? | Monitor SLA + workload |

**INTERVIEW CLOSE**
I do not assign ownership from the symptom. I prove the transaction, find the first failed handoff, make the smallest safe change, and verify that the patient-facing workflow still works after deployment.

# How I would move a change safely into production

In a real hospital LIS environment, the recommendation is not the finish line. A clinically correct idea can still become a production incident if it is built, tested, deployed, or monitored poorly. I would expect a controlled lifecycle with clear ownership and rollback.

| Phase | Required evidence |
|---|---|
| **1. Reproduce + baseline** | Defect reproduced; current-state transaction/log evidence captured |
| **2. Risk + ownership** | Patient-safety/downstream impact, accountable owner, and scope identified |
| **3. Build + document** | Non-production configuration change documented and versioned |
| **4. Integration + negative tests** | Expected path works; ambiguous/bad/error cases fail safely |
| **5. Regression + UAT** | Unaffected workflows pass; lab/application owner signs off |
| **6. Approve + rollback** | Deployment window, communication, and rollback criteria approved |
| **7. Production canary** | Small controlled transaction set verified end to end |
| **8. Post-live verification** | Queues, ACKs, filing, display, KPIs, and exceptions monitored |

## Exception handling: define what happens when the transaction is unsafe

| Condition | Safe design requirement |
|---|---|
| **No ACK / AE / AR** | Trace message control ID; route to interface exception process; reprocess only after cause is known |
| **MRN namespace ambiguous** | Do not silently file; follow defined patient-identity exception workflow |
| **Order / accession mismatch** | Prevent or hold auto-filing per local design; reconcile relationship before release |
| **Invalid chronology** | Exclude from KPI; investigate timestamp source/semantics before use |
| **Unknown test-code mapping** | Route to controlled mapping exception; never silently substitute another test |

## Validation matrix before go-live

| Test | Scenario | Pass condition |
|---|---|---|
| **VAL-01** | CBC with differential round trip | Correct LIS order + all expected components return |
| **VAL-02** | Critical potassium | Value/units/HH/final status preserved; notification workflow tested separately |
| **VAL-03** | Patient namespace | MRN + assigning authority/type relationship remains intact |
| **VAL-04** | Order/accession pairing | Correct placer/filler relationship remains intact |
| **VAL-05** | Timestamp sequence | Collection <= receipt <= final under defined semantics |
| **VAL-06** | Silent mapping + ACK/error path | CBCDIFF cannot silently become CBC; message control ID remains traceable |